

Discrete Mathematics¹

<http://lcs.ios.ac.cn/~znj/DM2017>

Naijun Zhan

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Outline

1 Sets and Functions

2 Cardinality, Diagonalization Argument

- Fix an universal set U . Set operations: union $\cup$, intersection $\cap$, complement $\bar{A}$.
- Set inclusion: $A \subseteq B$ iff for all $a \in A$ it holds $a \in B$. $A = B$ iff $A \subseteq B$ and $B \subseteq A$.
- Given a set S , the *power set* of S is the set of all subsets of the set S . The power set is denoted by $\mathcal{P}(S)$, or 2^S .
- The *Cartesian product* of sets $A_1, A_2, \dots, A_n$ is defined by:
 $A_1 \times \dots \times A_n := \{(a_1, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, \dots, n\}$.
- The *cardinality* of finite set A , denoted by $|A|$, is the number of its elements. The *principle of inclusion-exclusion*:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

TABLE 1 Set Identities.

<i>Identity</i>	<i>Name</i>
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cup C) = (A \cap B) \cup C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Let A and B be nonempty sets. A function $f: A \rightarrow B$ from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if $b \in B$ is assigned by f to the element $a \in A$. We say that

- A is the *domain* of f ,
- B is the *codomain* of f .
- If $f(a) = b$, we say that b is the *image* of a and a is a *preimage* of b .
- The *range*, or image, of f is the set of all images of elements of A .

Relation

Given two sets A and B ,

- a relation R between A and B is a subset of $A \times B$.
- a function from A to B must be a relation, but the inverse does not hold in general.

Let A and B be two sets. The function $f: A \rightarrow B$ is called

- *one-to-one*, or an *injection*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .
- *onto*, or a *surjection*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.
- *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto.

Let A , B , and C be three sets.

- Let $f: A \rightarrow B$ be bijective. The *inverse function* of f , denoted by f^{-1} , is the function that assigns to an element $b \in B$ the unique element $a \in A$ such that $f(a) = b$.
- Let $g: A \rightarrow B$ and let $f: B \rightarrow C$. The *composition* of the functions f and g , denoted $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a))$$

Let A , B , and C be three sets.

- Let $R \subseteq A \times B$, then the *inverse relation* of R , denoted by R^{-1} , is a subset of $B \times A$ such that $(b, a) \in R^{-1}$ iff $(a, b) \in R$, which is a relation between B and A .
- Let $R_1 \subseteq A \times B$ and $R_2 \subseteq B \times C$. The *composition* of the relations R_1 and R_2 , denoted $R_1 \circ R_2$, is defined by

$$(R_1 \circ R_2) = \{(a, c) \mid \exists b. (a, b) \in R_1 \wedge (b, c) \in R_2\}$$

Let A and B be two sets.

- For a function $f: A \rightarrow B$, and a set $D \subseteq A$, we use $f|_D: D \rightarrow B$ to denote the function f with domain restricted to the set D .
- A *partial function* f from a set A to a set B is an assignment to each element $a \in D \subseteq A$, called the *domain of definition* of f , of a unique element $b \in B$. We say that f is undefined for elements in $A \setminus D$. When $D = A$, we say that f is a total function.

Consider the set $U = 2^{AP}$ of all assignments. The semantic bracket is a function $\llbracket \cdot \rrbracket : PL \rightarrow 2^U$ defined by:

- $\llbracket p \rrbracket = \{\sigma \in U \mid p \in \sigma\},$
- $\llbracket \neg \varphi \rrbracket = \overline{\llbracket \varphi \rrbracket},$
- $\llbracket \varphi \rightarrow \psi \rrbracket = \overline{\llbracket \varphi \rrbracket} \cup \llbracket \psi \rrbracket.$

Is $\llbracket \cdot \rrbracket$ injective, surjective, or bijective?

Let $A_1, \dots, A_n$ and B be sets

- for any $f \in A_1 \times A_2 \times \dots \times A_n \rightarrow B$, there exists $f' \in A_1 \rightarrow (A_2 \rightarrow \dots (A_n \rightarrow B) \dots)$, such that for any $(a_1, \dots, a_n) \in A_1 \times \dots \times A_n$, it follows $f(a_1, \dots, a_n) = f'(a_1)(a_2) \dots (a_n)$.
- Conversely, for any $f' \in A_1 \rightarrow (A_2 \rightarrow \dots (A_n \rightarrow B) \dots)$, there exists $f \in A_1 \times A_2 \times \dots \times A_n \rightarrow B$, such that for any $(a_1, \dots, a_n) \in A_1 \times \dots \times A_n$, it follows $f'(a_1)(a_2) \dots (a_n) = f(a_1, \dots, a_n)$.

Outline

1 Sets and Functions

2 Cardinality, Diagonalization Argument

Let A and B be two sets.

- The sets A and B have the same cardinality if and only if there is a one-to-one correspondence from A to B . When A and B have the same cardinality, we write $|A| = |B|$.
- If there is a one-to-one function from A to B , the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$. Moreover, when $|A| \leq |B|$ and A and B have different cardinality, we say that the cardinality of A is less than the cardinality of B and we write $|A| < |B|$.

Countable set

A set that is either finite or has the same cardinality as the set of positive integers is called *countable*. A set that is not countable is called *uncountable*. When an infinite set S is countable, we denote the cardinality of S by $\aleph_0$. We write $|S| = \aleph_0$ and say that S has cardinality aleph null.

SCHRÖDER-BERNSTEIN THEOREM

If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.

Prove that

- 1 The union, intersection of countable sets is countable.
- 2 The set $\mathbb{N}^2$ is countable.
- 3 The set $\mathbb{Z}$ of integer numbers is countable.
- 4 The set $\mathbb{Q}$ of rational numbers is countable.
- 5 The set $\mathbb{N}^c$ with $c \in \mathbb{N}$ is countable.
- 6 The countable union of countable sets is countable ([exercise](#)).
- 7 The set $\mathbb{N}^*$ is countable ([exercise](#)).

Prove that

- 1 $|[0, 1]| = |(0, 1)| = |[0, 1)| = |(0, 1]|$.
- 2 $|(0, 1)| = |[1, \infty)|$.
- 3 $|[0, 1]| = |[0, k]| = |[0, \infty)| = |\mathbb{R}|$ (exercise).
- 4 $|\{0, 1\}^\omega| = |[0, 1]|$.
- 5 $|2^{\mathbb{N}}| = |\{0, 1\}^\omega|$.
- 6 $|2^{\mathbb{N}}| = |\{ f \mid f: \mathbb{N} \rightarrow \{0, 1\} \}|$ (exercise).

Cantor diagonalization argument

- The set $\mathbb{R}$ of real numbers is uncountable.
- For a set A , it holds: $|A| < |2^A|$.

- Prove $\mathbb{R}$ is uncountable.
- Prove $(\varphi_1 \rightarrow \psi_1) \vee (\varphi_2 \rightarrow \psi_2) \vdash (\varphi_1 \rightarrow \psi_2) \vee (\varphi_2 \rightarrow \psi_1)$.
- Prove $(\varphi \wedge \psi) \rightarrow \theta, \theta \rightarrow \chi, \psi \wedge \neg \chi \vdash \neg \varphi$.